



NGUYEN THANH DAT

Backend Engineer · NestJS / TypeScript · PHP / Laravel

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PROFESSIONAL SUMMARY

- Nearly 5 years of backend engineering across ERP, MES, healthcare, warehouse and e-commerce systems, mostly for Japanese clients — **PHP/Laravel for four of them**, including a **Python/Django to Laravel re-platform**, and for the past year sole owner of a **NestJS/TypeScript ERP** service end to end.
- **The seam is the specialty**: I keep a Laravel estate and a NestJS service agreeing in production. B2B and MES (PHP) write each day's business data into a shared sync database; **12 scheduled jobs and 3 RabbitMQ consumers** pull it across and recompute the ERP summaries.
- **Performance work is root-cause, not guesswork**: every result below traces to a query plan, a missing index or a composite index with the wrong column order, measured before and after.
- **Leading and owning**: as **Team Leader** I set the architecture, database design and technology stack for a **5-person backend team** from inception, and stepped into delivery management on a project that had lost three PMs in three months. Today I am the sole developer responsible for one production ERP service, which aggregates the B2B and MES systems beside it.

TECHNICAL SKILLS

Languages & Frameworks: PHP/Laravel (4 yrs) · TypeScript (2 yrs) · JavaScript (2 yrs) · NestJS (1 yr) · Vue 3 (2 yrs) · Tailwind · Lumen

Databases & Performance: MySQL (3 yrs) · PostgreSQL (2.5 yrs) · TypeORM · Eloquent · Schema Design · Indexing · Query & Index Tuning · Execution-Plan Analysis · Redis Caching Strategies

Architecture & Async: Modular Architecture · RESTful API Design · **Legacy System Migration (Django → Laravel)** · Queue-Based / Async Processing (RabbitMQ, AWS SQS) · Scheduled Batch Jobs · Cross-System Data Sync · OAuth2 / JWT (Passport) · SSO (Keycloak)

Infrastructure & Deployment: Docker · Nginx · Container Resource Isolation · Linux · AWS S3 · **SSL/TLS (Let's Encrypt, certbot)** · GitHub Actions (CI/CD) · Cloudflare Pages · **Staging Environment Setup** · **System Monitoring & Logging** · Git · GitHub · GitLab

Testing, Monitoring & Docs: Vitest · Unit & Feature Testing · Swagger · Laravel Telescope · Technical Documentation

Integrations: Stripe API · OpenAI API · Firebase Cloud Messaging · Pusher · Laravel Reverb · Google Maps API · SNS Login · Fax API · PDF Generation & Export

Leadership & Process: Technical Leadership (5-person backend team) · System & Database Design · Git Flow & Release Management · Sprint Planning & Scheduling · Code Review Process · AI-Assisted PR Review · Mentoring · Scrum · Jira · Redmine · Backlog · Trello

Working knowledge (side project): Go (Gin) · FastAPI · Bun

PROFESSIONAL EXPERIENCE

BEAR ACER COMPANY — Senior Backend Developer

Aug 2025 – Present | Japanese client | ERP service owned solo

- Sole developer on the ERP service — **NestJS with TypeORM** on the server and **Vue 3 with TypeScript** on the client, both mine. Inherited it and now own it end to end: architecture, features, frontend, performance and releases. It aggregates the B2B and MES PHP systems for daily/monthly reporting, support and email.
- **The client-facing monthly report: 26s → 250ms**. Replaced an on-the-fly aggregation over five million rows with a pre-computed rollup table, and rebuilt a composite index whose column order was wrong — **rows scanned 5M → 380K**.
- **Three CRM endpoints: 18.2s → 64ms, 8.8s → 60ms, 6.1s → 187ms**. The worst was a LEFT JOIN across ~18 tables (contacts 1M, contact info 2M rows); I split it into a fast main-record call plus a deferred contract call for the heavier data.
- **A multi-column filter: 9s → 80ms**. The filter combination had no index to land on. Diagnosing that was the work; adding the index it was missing took minutes.
- **The API was fast and the screen was still slow**. One report page painted 100 rows against roughly 100 columns — twelve months by eight figures each, plus totals and the yearly average — into a vxe-table, so the browser was the bottleneck once the query returned. Capped what the front end pulls per page and moved the columns onto horizontal lazy loading.
- **Slow-query alerting**: any query over 1s is logged and posted to the team's Discord channel, so a regression reaches the developers early instead of reaching the client.
- Batch jobs and the app server shared one container on an inherited codebase with no monitoring — docker stats showed the CPU/RAM contention after users reported slowdowns. Split batch into a dedicated worker container with hard limits (0.5 CPU / 512MB) and Node.js CPU self-throttling, stabilising the app server.
- Built the cross-system sync between the PHP estate and the ERP — a shared sync database, 12 scheduled batch jobs and 3 RabbitMQ consumers recomputing the summaries.
- **Designed and built the MES Billing module (Laravel)**: multi-case billing calculation, PDF invoice generation re-mapped on edit and re-download, multi-recipient email delivery, billing list, and role-based permissions.
- Built an **AI email-drafting feature**: it composes emails in the voice of the user sending them, matching their tone and writing style.

- Implemented real-time inventory updates with Redis, S3 and Laravel Reverb; wrote unit and feature tests for new work; ran Laravel Telescope on B2B/MES.

Tech Stack: NestJS, TypeORM, Vue 3, TypeScript, Tailwind, MySQL, PostgreSQL, RabbitMQ, Redis, Laravel Reverb, Laravel Telescope, Docker, Nginx, certbot, Vitest, Laravel, S3, Linux

SOTATEK COMPANY — Team Leader (Backend)

Mar 2025 – Aug 2025 | Japanese client | 16-member team, 5-person backend team

- **Team Leader for the 5-person backend team** on a hospital management system built from inception: I owned the system architecture, the database design and the technical solutions, and ran the team's Git flow.
- **The project had run through three project managers in three months** and was slipping. Without holding the title I stepped in on delivery — planning, timelines and release schedules for dev and QC — and focused on what was actually causing the issues.
- **Rebuilt the release process on a project carrying real delivery risk:** ad-hoc “dev-environment” releases became feature branches cut from main with sprint-scoped planning, so QC could plan test cases per sprint and nothing reached production without clearing QC first — avoidable issues fell and releases held their schedule.
- **Built the staging server that the new process needed** — the old flow had nowhere to verify a release before it shipped — and added **system monitoring and logging** so an issue could be investigated from evidence instead of guessed at.
- Raised scope risk before it became a bug: when a ticket the client wanted was likely to cause problems, the trade-off was explained and confirmed with them before it entered a sprint.
- Introduced a peer-review checklist, then **built an AI-assisted PR review step** once pull-request volume outgrew what one reviewer could cover, so review capacity stopped being the ceiling on quality. Mentored two developers through their onboarding.
- Found the slow path in appointment booking: the availability lookup across doctor schedules, shifts and existing appointments had no composite index to land on. Added one and restructured the query around how the screen actually loads its data. Also implemented OAuth via Laravel Passport.

Tech Stack: Lumen, Laravel, MySQL, Redis, Laravel Reverb, Docker, Nginx, S3, OAuth2/Laravel Passport, Firebase Cloud Messaging, Swagger, Linux

KOZOCOM COMPANY — Backend Developer (Middle-level)

Feb 2024 – Mar 2025 | Japanese client | 16-member team

- **Re-platformed a legacy Python/Django system onto Laravel and PostgreSQL** with a 16-member Scrum team — one product combining sales with warehouse and inventory management — under clean-code and clean-architecture practices.
- Carried the data across: **CSV import of ~2–3 million records per run, and export at comparable volume**, written as chunked insert() batches of 50 rows per worker rather than one model create per row. At that volume the risk is not speed but PHP running out of memory before the job finishes.
- Where the legacy behaviour was ambiguous I read the original Django implementation and **confirmed the intended business rule with the client** rather than guessing, so the rewrite followed how they actually run the business.
- Cut the read-path cost of the inventory list, which the client had raised as a complaint: eager-loaded the relations it had been resolving one row at a time (a textbook $N+1$), moved the list off paginate() so it stopped paying for a COUNT over the whole table on every request, and **re-indexed it** — the driving table held **1–2M rows and joined several others of the same order**, with nothing matching how the screen actually filtered. Afterwards the list opened straight away.
- Designed the schemas and business logic along the customer's operating flow: **goods-in, invoicing, shipping against a customer order, remaining-stock recalculation and business reporting**. The platform sells as well as stores — each sale's data flows back into the warehouse, where the warehouse costs are calculated from it.
- **Owned every PDF feature across both codebases**, inventory and e-commerce, from generation through to export.
- An **AWS SQS** queue handled batched data sync platform-wide, with invoice delivery by **fax through a third-party API**. Added feature and unit tests, wrote the module documentation, reviewed code and supported junior developers.

Tech Stack: Laravel, PostgreSQL, AWS SQS, Swagger, Redis, Pusher, Fax API, PDF Export, Nginx, Linux, Python/Django (legacy source)

FLYDINO TECHNOLOGY COMPANY — Junior Full-Stack Developer

Apr 2023 – Feb 2024 | Japanese & Vietnamese clients | 8–18 member teams

- **Full-stack on three client platforms** — AI skin analysis, driver booking, and project management with recruitment: Laravel on the backend for all three, and the frontend on two of them in **Vue 3 with TypeScript** and Tailwind, designing the database structures for the new features.
- **Integrated ChatGPT-4** into a skin-analysis flow: an uploaded facial photo returns a skin-condition assessment that feeds a personalised product-recommendation step.
- On the project-management and recruitment platform (18-member team): project, task and contract tracking in the shape of Backlog or Asana, **wired to a hiring flow so a project short of people can source candidates — onsite, remote or freelance — without leaving the system**. Integrated Keycloak SSO, SNS login and S3/FTP storage, and **re-indexed the slow list queries** behind the project and task screens.

Tech Stack: Laravel, Vue 3, TypeScript, Tailwind, MySQL, Swagger, S3, Keycloak, Pusher, Google Maps API, Nginx

NINE PLUS SOFTWARE DEVELOPMENT COMPANY — Backend Developer (Fresher → Junior)

Oct 2021 – Apr 2023 | Japanese client | 12–20 member teams

- **Promoted from Fresher to Junior** over 18 months, moving from assigned tasks to **owning features end to end** — database schema, business logic and API — with a senior reviewing the work before it merged.
- Implemented features for an HR management system — schemas and business logic for the recruitment and HR modules — then across three platforms: two online-retail systems and a **product-review platform**. Product, order and order-item management throughout, plus **Stripe** on the two retail systems: I built the Laravel side and the queued processing that let the

endpoint answer immediately, **working with a senior** who guided me through signature verification, idempotent handling of retried events, and the refund, failure and reconciliation paths.

Tech Stack: Laravel, MySQL, S3, Stripe API, Docker, Nginx

SELECTED PROJECTS

B2B ERP & MES Platform

Aug 2025 – Present | Japanese client | ERP service owned solo

- Enterprise ERP and MES covering accounting, CRM, HR, billing, orders, shipping and inventory. The ERP service is mine end to end; the B2B and MES systems it aggregates from are PHP, and I contribute there too.

Tech Stack: NestJS, TypeORM, Vue 3, TypeScript, Tailwind, MySQL, PostgreSQL, RabbitMQ, Redis, Docker, Nginx, Vitest

Hospital Management System

Mar 2025 – Aug 2025 | Japanese client | 16-member team, 5-person backend | built from inception

- Appointment booking against doctor schedules and shifts, plus course, shift and contract management, OAuth via Laravel Passport and push notifications. **As Team Leader I owned the architecture, database design and stack for the 5-person backend team.**

Tech Stack: Lumen, Laravel, MySQL, Redis, Laravel Passport, Firebase Cloud Messaging, Swagger, Docker, Nginx

Sales, Warehouse & Inventory Platform

Feb 2024 – Mar 2025 | Japanese client | 16-member team

- A legacy Python/Django system re-platformed onto Laravel and PostgreSQL**, combining sales and warehouse in one product: purchasing, selling and store management, goods-in, invoicing, shipping against customer orders, remaining-stock recalculation and business reporting, with each sale feeding back into the warehouse costing. PDF documents and export, fax delivery through a third-party API, and an AWS SQS queue carrying batched data sync platform-wide.

Tech Stack: Laravel, PostgreSQL, AWS SQS, Redis, Swagger, Pusher, Fax API, PDF Export, Nginx, Linux, Python/Django (legacy source)

AI Skin Analysis & Booking Platform

Dec 2023 – Feb 2024 | Japanese client | 8-member team

- ChatGPT-4 reads an uploaded facial photo**, returns a skin-condition assessment, and feeds a personalised product-recommendation flow. Built the service-booking, store-management and authentication/authorization features end to end.

Tech Stack: Laravel (MVC), OpenAI API, JavaScript, Tailwind, MySQL, Nginx

Project Management & Recruitment Platform

Jun 2023 – Dec 2023 | Japanese client | 18-member team

- Project, task and contract modules with role-based authorization, in the shape of Backlog or Asana — **wired to a recruitment flow so a project short of people can source candidates for onsite, remote or freelance engagement inside the same system.** Keycloak SSO, SNS authentication, and storage across S3 and FTP.

Tech Stack: Laravel (RESTful API), Vue 3, TypeScript, Tailwind, MySQL, Swagger, S3, Keycloak, Pusher, Nginx

Driver Booking Platform

Apr 2023 – Jun 2023 | Vietnamese client | 8-member team

- Real-time distance and fare calculation between pickup and destination via the **Google Maps API**, plus driver booking, ratings, and driver/user management.

Tech Stack: Laravel (RESTful API), Vue 3, TypeScript, Tailwind, MySQL, Google Maps API, Nginx

E-commerce, Payments & Review

Mar 2022 – Jan 2023 | Japanese client

- Online retail** — product, order and order-item management, CSV import/export, and online payment. Oct 2022 – Jan 2023, 9 members.
- Product review platform** — a publishing platform for reviewers: review posts on products, moderation, search and filtering, and ratings. Aug – Oct 2022, 7 members.
- Multi-seller storefronts** — seller registration, product, order, order-item and store management, and the payment and shipping workflow. Mar – Aug 2022, 12–20 members.
- Integrated Stripe checkout on the storefronts and the online-retail platform, SNS login on the storefronts and the review platform, and Redis caching the catalog reads.

Tech Stack: Laravel (RESTful API & MVC), MySQL, Stripe API, Redis, Docker, Nginx, S3, JavaScript, Bootstrap, Pusher, Swagger

PERSONAL PROJECTS

MIA-POS — Self-funded Side Project (in testing)

Ongoing | Preview build: mia-store-remote.pages.dev

- Self-funded POS / online-store platform split by language behind **Nginx**, with certbot issuing the TLS certificates: Laravel for auth, admin and payments; **Go with Gin** for stores, inventory and the log worker; Python with **FastAPI** for the AI features; three **Vue 3** TypeScript apps in a Bun workspace.
- The services still share one database — a modular monolith, a deliberate trade-off at this stage. CI/CD on GitHub Actions.

Tech Stack: Laravel, Go, Gin, FastAPI, PostgreSQL, PgBouncer, Redis, RabbitMQ, Soketi, Vue 3, TypeScript, Tailwind, Bun, Nginx, certbot, Cloudflare Pages, GitHub Actions

EDUCATION & LANGUAGES

FPT Polytechnic — Software Application | Jul 2019 – Dec 2021

Languages: Vietnamese — native · English — reading and writing (technical docs and specifications).